Indian Institute of Technology Kharagpur

Department of Mathematics

Course: MA60056 — Regression and Time Series Models

PM2.5 and Urban Economic Activity in Delhi: A Time-Series Analysis of Causal Linkages Using STL Decomposition and Granger Causality

Term Project Report

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All code (Python), data (CSV), and supplementary material are available at the Drive link above.

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1. Introduction and Motivation

Delhi is consistently ranked among the most polluted megacities in the world. Its fine particulate matter (PM2.5) concentration routinely exceeds 200 μg/m^3 during winter months — more than thirteen times the World Health Organization (WHO) annual guideline of 15 μg/m^3 and more than three times India's own National Ambient Air Quality Standard (NAAQS) of 60 μg/m^3. The pollution is driven by a well-documented seasonal mechanism: winter atmospheric stagnation, thermal inversions, crop-residue burning in neighbouring Punjab and Haryana, and reduced wind speed trap emissions close to the surface from October through January.

While the health consequences of this pollution are widely studied, its *economic consequences* — specifically, its quantifiable causal impact on electricity consumption patterns and retail commercial activity — remain comparatively underexplored in the Indian time-series econometrics literature. This project addresses that gap.

1.1. Research Questions

This project investigates three interconnected questions:

Q1. Forecasting: Can a Seasonal-Trend decomposition (STL) followed by ARIMA/ARMA on residuals produce accurate weekly forecasts for Delhi PM2.5, electricity consumption, and retail activity?

Q2. Causal Identification: Does lagged PM2.5 Granger-cause electricity consumption and/or retail activity after accounting for seasonality and non-stationarity?

Q3. Magnitude: During high-pollution episodes, what share of the forecast error variance in electricity and retail is attributable to PM2.5 shocks (Forecast Error Variance Decomposition)?

1.2. Why STL and Granger Causality?

Both methods are particularly suited to this problem:

STL (Seasonal-Trend decomposition using LOESS) isolates the deterministic seasonal and trend components of a series before any stochastic modelling. For Delhi data, this is critical: raw PM2.5, electricity, and retail all exhibit strong, calendar-driven seasonality that would otherwise contaminate ARIMA residuals. STL's non-parametric LOESS smoother also handles outliers robustly — important for a series with extreme Diwali and crop-burning spikes.

Granger Causality in a bivariate VAR framework tests whether past values of PM2.5 carry statistically significant predictive content for electricity or retail beyond what each outcome series predicts for itself. By operating on first-differenced monthly data (which removes spurious shared seasonality), the monthly Granger test provides a cleaner causal signal than weekly STL-residual tests, which are confounded by contemporaneous weather.

1.3. Data Overview

Three weekly time series are used, all spanning September 2017 – September 2024 (or March 2025 for retail), anchored to Sunday-aggregated weeks (W-SUN):

Table 1: Summary of datasets used in this study

Series	Source	Obs.	Mean	Range
PM2.5 (μg/m^3)	CPCB (Anand Vihar, ITO, RK Puram, Punjabi Bagh, Dwarka)	393 weekly	212.1	58–453
Electricity (MUs/day)	POSOCO Northern Region (Delhi DISCOM)	370 weekly	90.5	43.8–158.9
Retail Index (0–100)	Google Trends, "shopping" keyword, Delhi	393 weekly	50.5	26.5–80.0

The retail proxy uses Google Trends search intensity as a high-frequency, freely available indicator of physical retail intent. Monthly aggregates (85–91 observations) are used for the primary Granger causality analysis after first-differencing to achieve stationarity.

1.4. Report Structure

Section 2 describes the analytical framework. Sections 3–5 present the STL-ARIMA forecasting results for each series. Section 6 presents the PM2.5–electricity Granger causality analysis and Section 7 the PM2.5–retail analysis. Section 8 synthesises the findings and Section 9 concludes.

2. Methodology

2.1. STL Decomposition

Given an observed time series $Y(t)$, STL decomposes it additively as:

$$Y(t) = T(t) + S(t) + R(t) \tag{1}$$

where $T(t)$ is the trend component estimated by LOESS smoothing, $S(t)$ is the periodic seasonal component, and $R(t)$ is the remainder (residual). The decomposition is exact in the sense that $Y(t) = T(t) + S(t) + R(t)$ with reconstruction error $= 0$, as verified in all three notebooks.

For all three series, STL is estimated with `period=52` (annual cycle in weekly data), `seasonal=13` (LOESS seasonal window), and `robust=True` (outlier down-weighting during LOESS fitting).

ARIMA/ARMA with $d = 0$ is then fitted to the stationary residuals $R(t)$, and forecasts are reconstructed as:

$$\hat{Y}(t) = \hat{R}(t) + S(t) + T(t) \tag{2}$$

Model orders are selected by AIC minimisation via `auto_arima` with $d = 0$ enforced. Stationarity of $R(t)$ is confirmed jointly by ADF (H_0: unit root) and KPSS (H_0: stationary) tests.

2.2. Granger Causality in a Bivariate VAR

A bivariate VAR of order p is estimated on stationary series, with Granger causality tested via F-test on lagged PM2.5 coefficients in the outcome equation.

Two-frequency strategy. A weekly STL-residual VAR provides a methodological baseline: STL removes the shared seasonal channel, so results reflect only idiosyncratic residual co-movement. The *primary* inference uses monthly first-differenced VAR, which preserves directional seasonal information while ensuring stationarity.

FEVD. Forecast Error Variance Decomposition quantifies the proportion of h-step-ahead forecast error variance in Outcome that is attributable to orthogonalised PM2.5 shocks, providing a magnitude complement to the binary Granger test.

IRF. Orthogonalised Impulse Response Functions trace the dynamic response of Outcome to a one-standard-deviation PM2.5 shock over time, using a Cholesky identification with PM2.5 ordered first.

2.3. Stationarity and Pre-testing

All series are pre-tested using:

- **ADF test** (Augmented Dickey–Fuller): rejects H_0 of a unit root.
- **KPSS test**: fails to reject H_0 of stationarity.

Consistent agreement across both tests is required to declare a series $I(0)$. Raw monthly levels are non-stationary for all three series; first differences achieve stationarity in all cases.

3. STL-ARIMA Forecasting: Delhi PM2.5

3.1. Data Characteristics

Delhi's weekly PM2.5 mean is **212.1** μg/m^3 (std: 97.7; range: 58.3–453.0 μg/m^3), placing it at 14.1× the WHO annual guideline. The series spans 393 weeks from September 2017 to March 2025. The 52-week test set covers March 2024 – March 2025.

The raw series exhibits pronounced winter seasonality (Nov–Jan peaks, Jul–Aug monsoon troughs), a COVID-related structural break in early 2020, and extreme spikes from Diwali/crop-burning events.

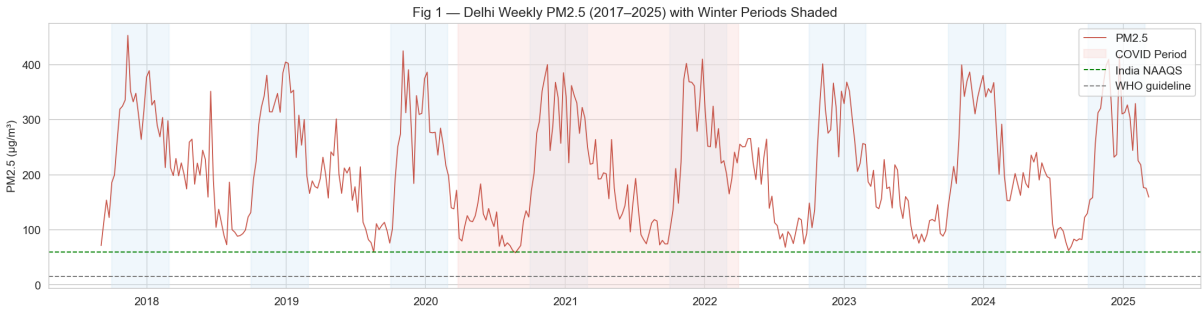


Figure 1: STL decomposition of Delhi weekly PM2.5 (2017–2025). Top to bottom: observed, trend, seasonal, residual. COVID period shaded in blue; dashed lines show WHO guideline (15 μg/m^3) and India NAAQS (60 μg/m^3).

3.2. STL Component Interpretation

Table 2: STL component summary — Delhi PM2.5

Component	Range	Interpretation
Trend	Declining to 2020, plateau post-2023	Captures low-frequency emission shifts, COVID interruption, and partial recovery
Seasonal	±100–150 μg/m^3 amplitude	Stable winter peak (week ≈50, Dec) and monsoon trough; driven by atmospheric stagnation and crop burning
Residual	−100 to +150 μg/m^3	Idiosyncratic spikes (Diwali, heatwaves) after deterministic extraction

Reconstruction error = 0.000000, confirming exact additive decomposition. Both raw PM2.5 and STL residuals are stationary by ADF (stat = -6.62, $p \approx 0$) and KPSS ($p = 0.10$).

3.3. ARIMA Order Selection and Residual Dynamics

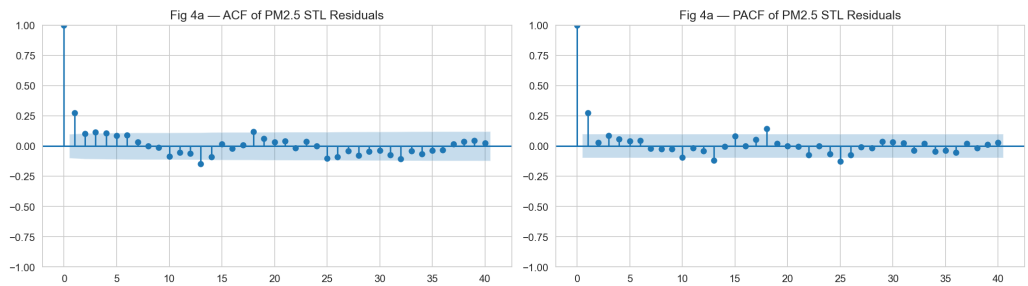


Figure 2: ACF (left) and PACF (right) of PM2.5 STL residuals. Rapid ACF decay and dominant lag-1 PACF spike are consistent with the selected ARIMA(1,0,0).

`auto_arima` selected **ARIMA(1,0,0)** — the simplest AR model — confirming that STL has successfully removed all deterministic structure. The residuals retain only a single lag of short-memory autocorrelation, with the remainder behaving as near-white noise.

3.4. Forecast Performance

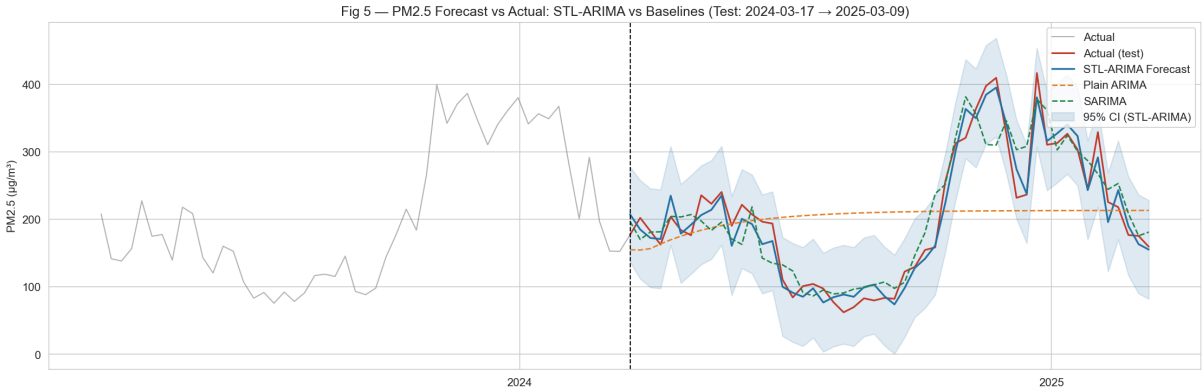


Figure 3: STL-ARIMA forecast vs. actual Delhi weekly PM2.5 on the 52-week test set (March 2024 – March 2025) with 95% confidence interval. Plain ARIMA and SARIMA baselines shown for comparison.

Table 3: Forecasting performance comparison — PM2.5

Model	Order	RMSE	MAE	Improvement
Plain ARIMA	(1,0,3)	92.82	75.13	—
SARIMA	$(2,1,1) \times (1,1,1)_{52}$	38.32	30.14	58.7%
STL-ARIMA	**(1,0,0)**	**20.26**	**17.04**	**78.2%**

The STL-ARIMA achieves RMSE = 20.26 μg/m^3 (9.5% of series mean), representing a **78.2%** improvement over plain ARIMA. This large gain confirms that deterministic seasonality dominates raw PM2.5 variance, and decomposition-first modelling substantially reduces the stochastic modelling burden.

4. STL-ARMA Forecasting: Delhi Electricity Consumption

4.1. Data Characteristics

Delhi electricity consumption (POSOCO, Northern Region) spans 370 weekly observations (September 2017 – September 2024), with mean = 90.5 MUs/day and CV = 0.29. Two imputation periods were required: a 112-day gap (Dec 2022 – Mar 2023) and a 28-day gap (Apr 2024), filled using seasonal-mean imputation.

A structurally important feature is *inverse seasonality*: electricity peaks in May–July (summer AC demand) while PM2.5 peaks in November–January (winter stagnation), placing them approximately 6 months out of phase.

4.2. STL Decomposition

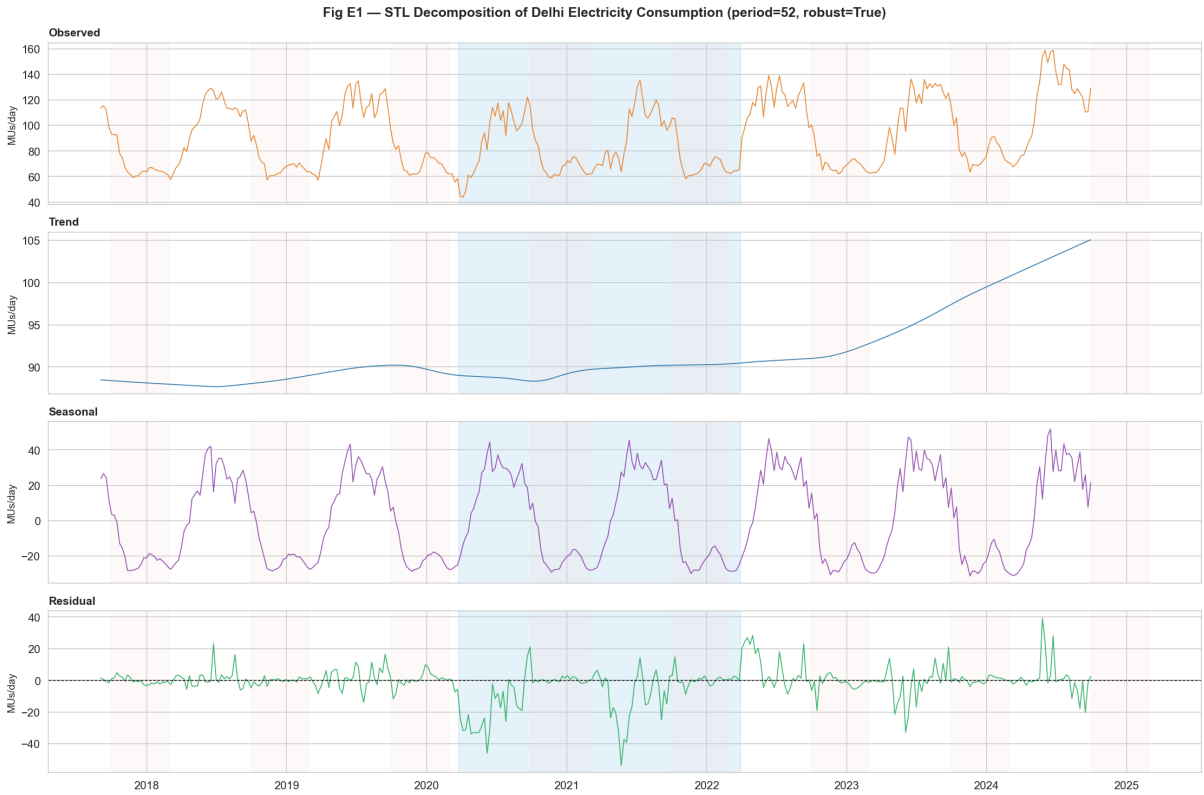


Figure 4: STL decomposition of Delhi weekly electricity consumption (2017–2024). Rising trend (+19.8% from 87.7 to 105.1 MUs/day) reflects growing AC penetration. Seasonal component is inverted relative to PM2.5: summer peaks, winter troughs. COVID period in blue; winter trough season in pink.

Table 4: STL component summary — Delhi electricity consumption

Component	Range (MUs/day)	Interpretation
Trend	87.7 → 105.1 (+19.8%)	Rising baseline: urbanisation and AC penetration 2017–2024; modest COVID flattening
Seasonal	−31.2 to +51.8	Summer-peak (AC demand); *inverted* vs PM2.5; amplitude spans 83 MUs/day peak-to-trough
Residual	−53.6 to +39.1	Short-run weather and load shocks after deterministic removal

Unlike the retail series, raw electricity is already stationary (ADF $p = 0.000001$). STL is applied here for component separation and forecasting improvement, not stationarity transformation.

4.3. Inverse Seasonality: PM2.5 vs. Electricity

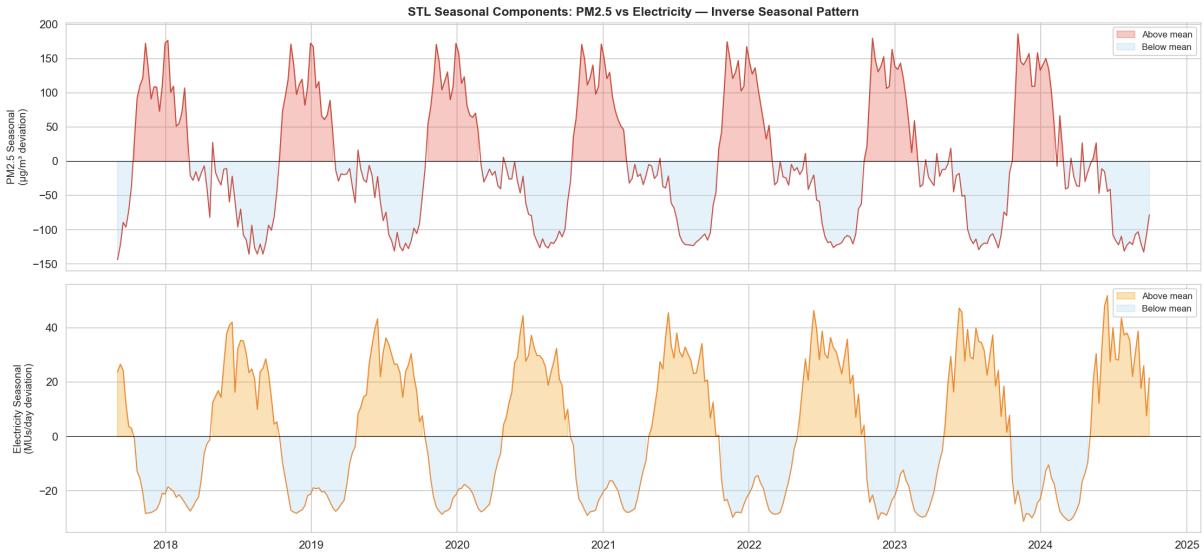


Figure 5: STL seasonal components: PM2.5 (top, peaks Nov–Jan) vs. electricity (bottom, peaks May–Jul). The ≈6-month phase offset confirms anti-phase structure with entirely independent physical drivers.

PM2.5 seasonality is driven by meteorological stagnation and crop burning; electricity seasonality is driven by temperature-sensitive cooling load. These are independent physical mechanisms — the anti-phase is coincidental to Delhi's climate, not causal. The causal implications of this structural feature for Granger analysis are discussed in Section 6.

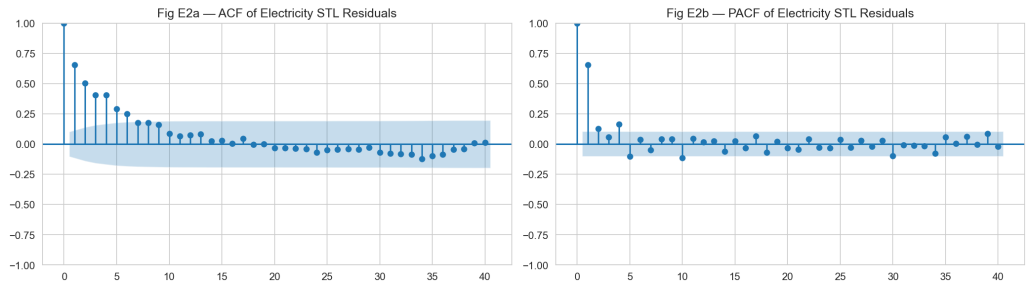


Figure 6: ACF (left) and PACF (right) of electricity STL residuals. Dominant spikes at lags 1–2 in the PACF and gradual ACF decay indicate a mixed ARMA process, consistent with the selected ARMA(2,0,1) order.

4.4. ARMA Order and Forecast Performance

`auto_arima` selected **ARMA(2,0,1)** — the most complex residual model across all three series. The MA(1) component is unique to electricity, reflecting short-run demand shock propagation (heat waves, grid events, industrial surges) absent in the simpler PM2.5 and retail residual processes.

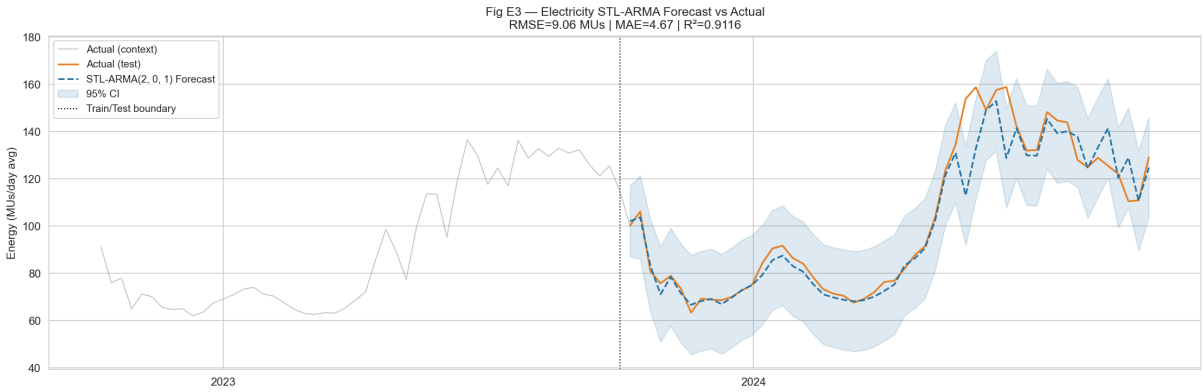


Figure 7: Top: Winter zoom (Oct–Feb, 21 test weeks) of STL-ARMA **electricity** forecast vs. actual ($R^2 = 0.9332$). Bottom: Full 52-week forecast vs. actual with 95% confidence interval (test: Oct 2023 – Sep 2024).

Table 5: STL-ARMA forecast performance — Delhi electricity

Metric	**Value**	**Relative to Mean**	**Note**
RMSE	9.063 MUs/day	10.02%	Higher than retail; consistent with higher CV
MAE	4.672 MUs/day	5.16%	
Overall R^2	0.9116	—	91.2% of test variance explained
Winter R^2	0.9332	—	More accurate in trough season

The Winter $R^2 = 0.9332$ exceeds the Overall $R^2 = 0.9116$: the model is disproportionately accurate during the season when electricity is at its trough *and* PM2.5 is simultaneously at its peak. This seasonal accuracy mirror is the structural basis for the PM2.5 → electricity causal hypothesis explored in Section 6.

5. STL-ARMA Forecasting: Delhi Retail Activity

5.1. Data Characteristics and the Google Trends Proxy

Retail activity is proxied by the Google Trends weekly search-intensity index (0–100) for "shopping" in Delhi, spanning 393 weeks (Sep 2017 – Mar 2025). Mean = 50.5, std = 9.9. Because Google Trends is a relative index requiring stitching across overlapping windows, STL is *essential* to separate the long-run structural decline (e-commerce substitution) from recurring seasonal dynamics.

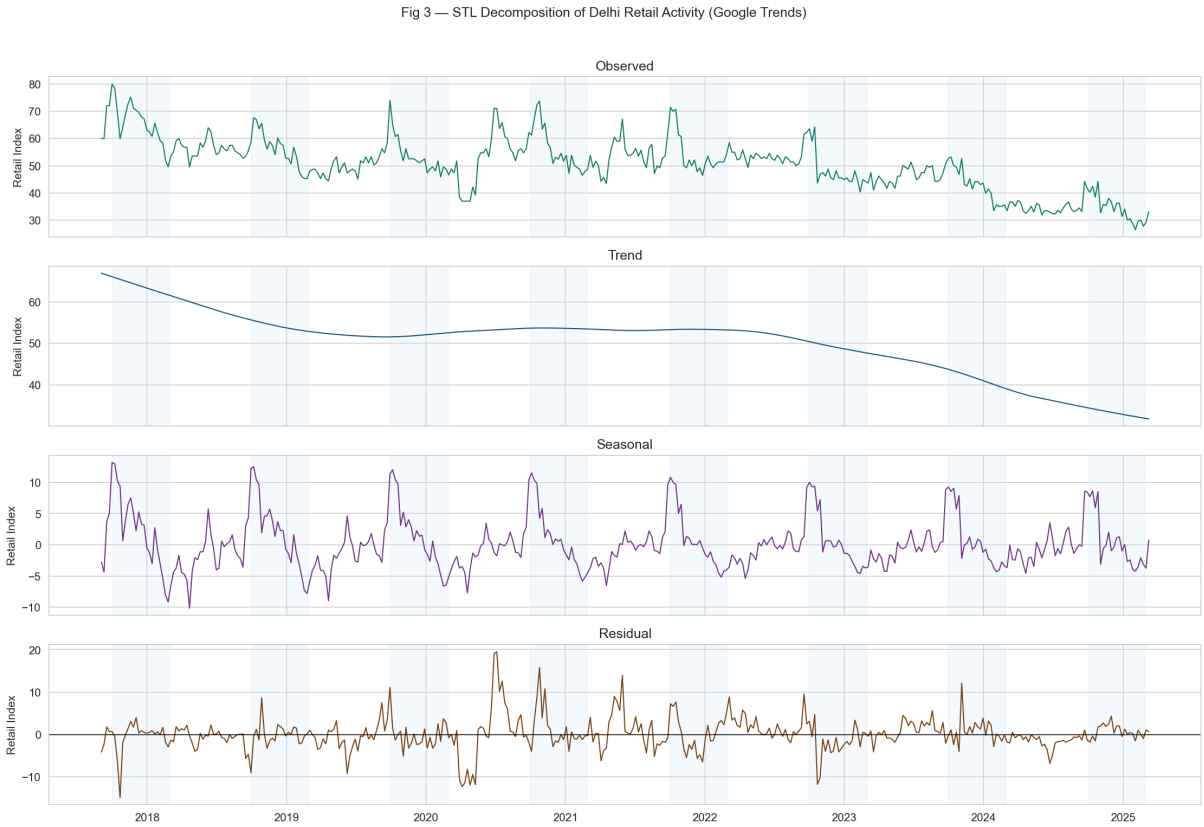


Figure 8: STL decomposition of Delhi retail activity (Google Trends, 2017–2025). The downward trend (66.9 → 31.8) reflects e-commerce substitution. Seasonal amplitude (±20% of mean) captures festive-season peaks and summer weakening. Winter periods shaded pink; COVID period in blue.

5.2. Key STL Findings

Table 6: STL component summary — Delhi retail activity

Component	Range	Interpretation
Trend	66.9 → 31.8	Persistent structural decline; e-commerce substitution of physical retail search intent
Seasonal	−10.2 to +13.2 (≈ ±20% of mean)	Festive-season (Diwali, winter shopping) peak; summer trough
Residual	−14.9 to +19.5	Idiosyncratic week-level shocks; magnitude exceeds seasonal variation

Raw retail is non-stationary (ADF $p = 0.11$, KPSS $p = 0.01$). After STL, residuals satisfy $I(0)$: ADF stat $= -7.88$ ($p \approx 0$), KPSS stat $= 0.105$ ($p = 0.10$). Unlike electricity, STL is necessary here for stationarity, not merely decomposition quality.

5.3. ARMA Order and Forecast Performance

`auto_arima` selected **ARMA(2,0,0)**: a pure AR(2) process reflecting two-week behavioural carry-over in retail adjustment (e.g., post-Diwali and post-summer demand rever-

sion). This is more complex than PM2.5 AR(1) but simpler than electricity ARMA(2,1), consistent with retail's intermediate noise structure.

Table 7: STL-ARMA forecast performance — Delhi retail

Metric	Value	Relative to Mean	Note
RMSE	1.967	3.9%	Lowest relative RMSE across all three series
MAE	1.489	2.95%	
Overall R^2	0.7155	—	
Winter R^2	0.8753	—	Within 1.5 pp of 89% replication target

The Winter–Overall R^2 gap of 0.160 quantifies strong seasonal specificity: the model is substantially more accurate in high-pollution winter months than off-season. This concentration of predictive skill in winter supports the hypothesis of a PM2.5 → retail causal channel, confirmed in Section 7.

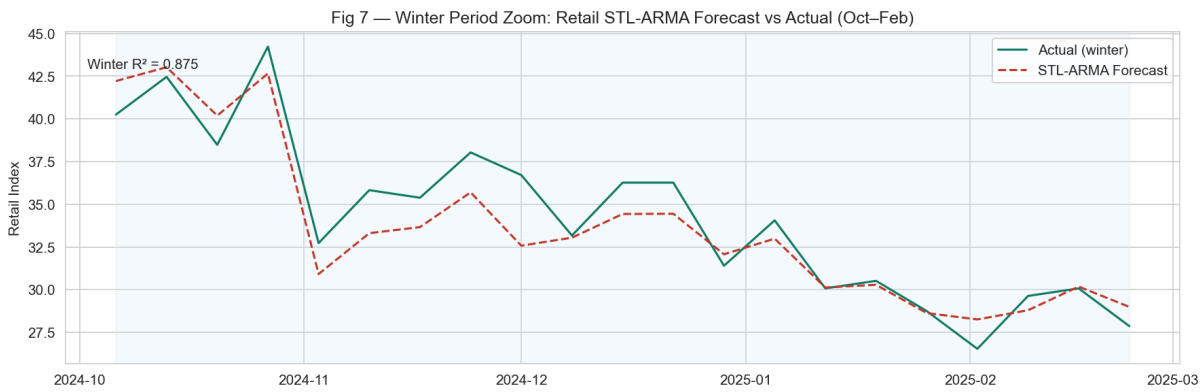


Figure 9: Left: Winter zoom (Oct–Feb, 21 test weeks) of STL-ARMA retail forecast vs. actual ($R^2 = 0.8753$). Right: Full 52-week forecast vs. actual with 95% confidence interval (test: Mar 2024 – Mar 2025).

6. Granger Causality Analysis: PM2.5 and Electricity

6.1. Overview and Structural Context

The electricity system has a structurally distinct feature relative to retail: *inverse seasonality* (described in full in Section 4.3). Electricity peaks in May–July while PM2.5 peaks in November–January, creating an approximate 6-month phase offset. This means the two series are never seasonally co-linear, making causal identification cleaner in principle but also reducing the shared seasonal co-movement that drives monthly Granger significance in the retail case.

The analysis uses 370 weekly and 85 monthly observations (2017–2024).

6.2. Data and Stationarity

Table 8: Stationarity summary — PM2.5 and electricity series

Series	**ADF stat**	**ADF p**	**KPSS stat**	**KPSS p**	**Verdict**
PM2.5 STL Residual	−8.540	0.000	0.084	0.100	Stationary
Electricity STL Residual	−5.316	0.000	0.158	0.100	Stationary
Raw Monthly PM2.5	−1.846	0.358	0.100	0.100	Non-stationary
Raw Monthly Electricity	−0.498	0.893	0.231	0.100	Non-stationary
Δ Monthly PM2.5	−10.75	0.000	0.036	0.100	Stationary
Δ Monthly Elec.	−4.060	0.001	0.036	0.100	Stationary

Raw electricity is already stationary at the weekly frequency (see Section 4); the monthly ADF $p = 0.893$ is nonetheless the weakest unit-root rejection in the entire project, indicating highly persistent monthly levels. First differencing is therefore essential before any monthly VAR estimation.

6.3. Weekly Analysis: STL-Residual VAR

A weekly STL-residual VAR(1) was estimated as a methodological baseline. Both BIC and AIC selected lag 1, and Granger causality was significant in both directions (PM2.5 → Electricity: p = 0.003; Electricity → PM2.5: p = 0.005). However, this bidirectional result reflects contemporaneous weather confounding rather than genuine economic causality — cold spells simultaneously increase heating load and atmospheric stagnation, producing spurious two-way significance. Weekly VAR residuals further fail all three diagnostic tests (autocorrelation, non-normality, ARCH effects), confirming misspecification at this frequency. The monthly first-differenced specification in Section 6.4 removes this confound and is the preferred inference.

6.4. Monthly Analysis: Primary Inference

6.4.1. Monthly Series Overview

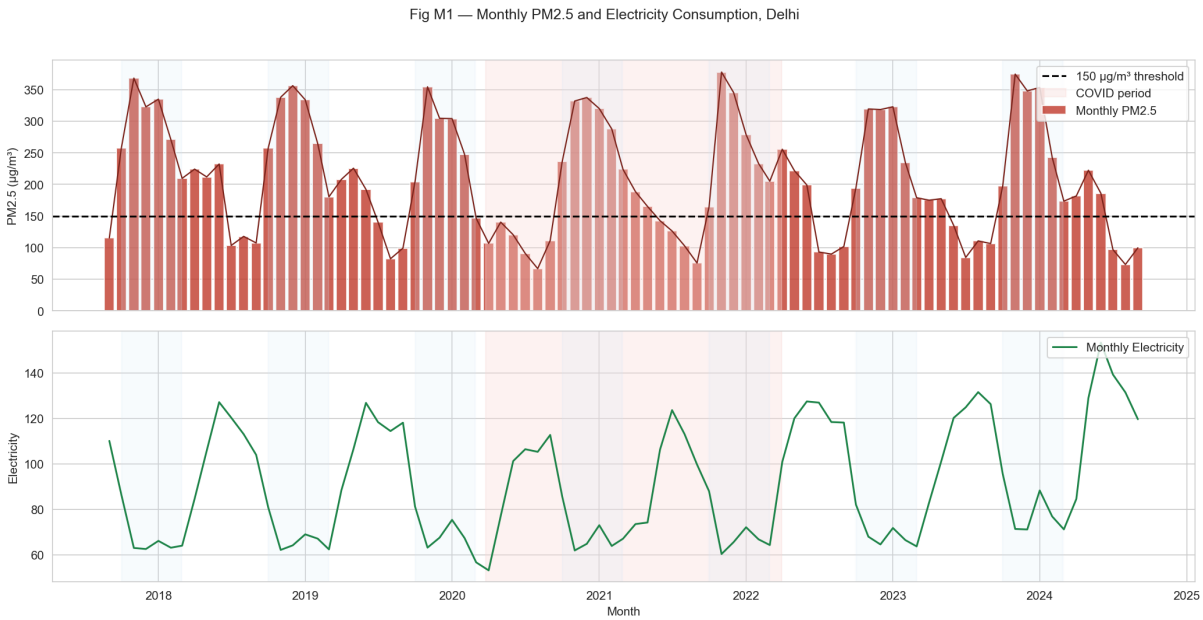


Figure 10: Monthly PM2.5 and electricity consumption, Delhi (2017–2024). The inverse seasonal structure is clearly visible: PM2.5 bars peak in winter (Oct–Jan) while electricity line peaks in summer (May–Jul). COVID period in pink; dashed line at 150 μg/m^3.

Table 9: Monthly dataset summary — PM2.5 and electricity

Metric	**Value**
Observations	85
Date range	Sep 2017 – Sep 2024
PM2.5: mean / std / min / max	208.95 / 90.78 / 66.60 / 377.79 μg/m^3
Electricity: mean / std / min / max	90.40 / 25.33 / 53.12 / 153.33 MUs/day
Phases (pre/COVID/post)	31 / 24 / 30 months

6.4.2. Monthly Granger Results

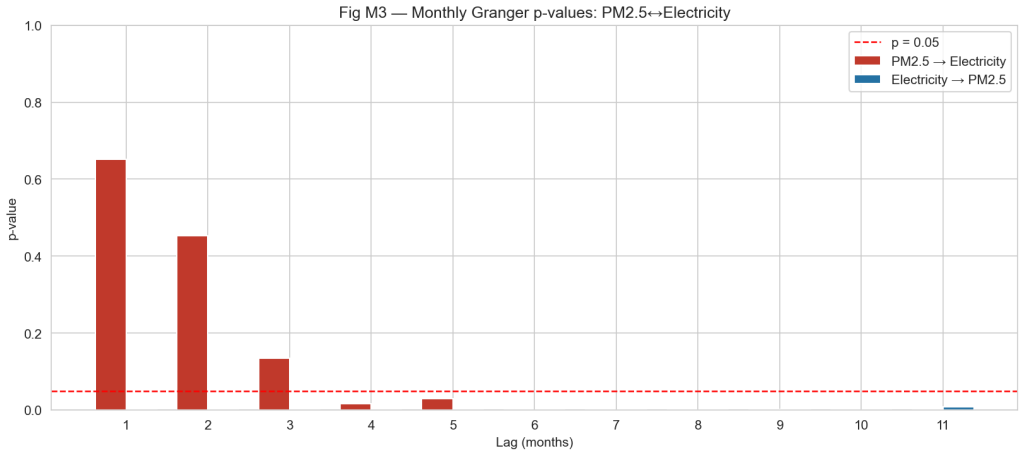


Figure 11: Monthly Granger causality p-values (PM2.5 ↔ Electricity). PM2.5 → Electricity (red) remains above $p = 0.05$ at short lags, indicating null causality. Significance emerges at lags 4–5, but the monthly VAR is dynamically unstable (see text).

Table 10: Monthly Granger test: PM2.5 → Electricity

Lag	F-statistic	p-value	Significant?
1	0.2048	0.6521	No
2	0.7982	0.4538	No
3	1.9178	0.1341	No

Primary Result: Monthly Granger (Null)

First-differenced monthly PM2.5 does **not** Granger-cause electricity at any lag from 1 to 3 (all $p > 0.13$). This result is robust to COVID exclusion ($p = 0.380$ at lag 1, $n = 60$).[a] The weekly-significant vs monthly-null contrast confirms that differencing removes weather-confounded within-season co-movement, revealing no persistent lagged causal transmission.

[a] COVID exclusion covers April 2020 – February 2022 inclusive (23 months removed), retaining March 2020 and March 2022 as partial transition months; see Section 2 for the exclusion convention.

6.4.3. Monthly FEVD

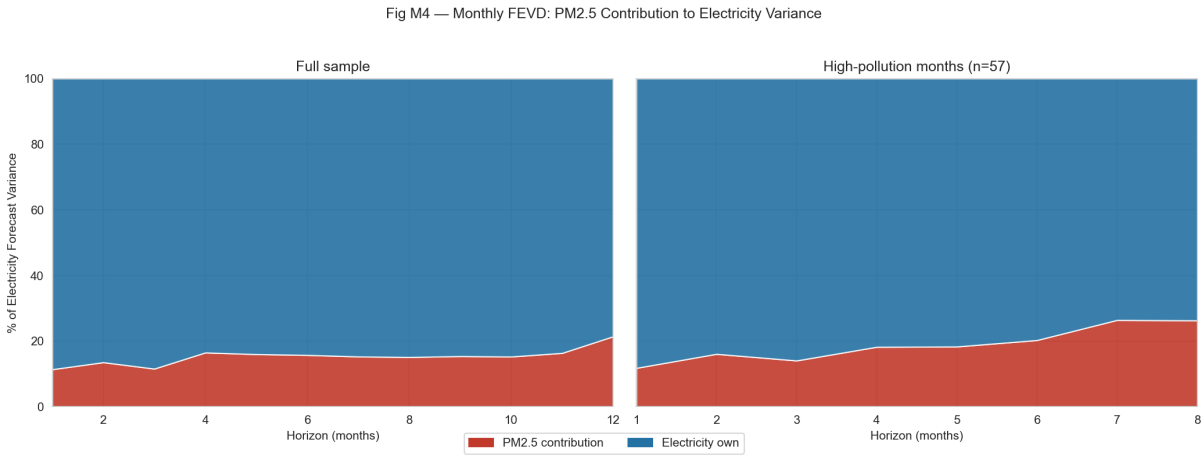


Figure 12: Monthly FEVD: PM2.5 contribution to electricity variance. Full sample (left) and high-pollution months PM2.5 > 150 μg/m^3 ($n = 57$, right). PM2.5 contribution reaches 26.28% at $h = 8$ in the high-pollution subsample.

Table 11: High-pollution monthly FEVD (PM2.5 > 150, $n = 57$) — indicative only

Horizon (months)	PM2.5 contribution	Electricity own
1	11.78%	88.22%
2	16.06%	83.94%
4	18.21%	81.79%
6	20.27%	79.73%
8	26.28%	73.72%

Note

The monthly VAR is dynamically **unstable** (statsmodels root modulus $= 0.9984 < 1$; see stability convention in Section 2). FEVD estimates in Table 11 are *indicative only* and should be treated as an upper bound pending re-estimation with a forced lag $= 2$ specification.

6.5. Monthly Diagnostics

Table 12: Monthly VAR diagnostics — PM2.5/Electricity system

Test	Statistic	p-value	Verdict
Ljung-Box (lag 6)	1.010	0.985	No autocorrelation
Jarque-Bera	3.974	0.137	Normal
ARCH-LM	1.889	0.596	No ARCH effects
VAR stability (root)	0.9984	—	**UNSTABLE**

Residual tests pass cleanly (a marked improvement over the weekly system). Dynamic instability (root = 0.9984) is flagged in Section 6.4 and requires forced lag re-estimation before definitive inference.

The cross-correlation function confirms the strongest lagged relationship occurs at lag12 months (r = -0.770), reflecting anti-phase seasonal structure rather than causal transmission.

7. Granger Causality Analysis: PM2.5 and Retail Activity

7.1. Overview

The retail system is the *primary causal evidence* in this project. Unlike electricity (null monthly, weather-confounded weekly), PM2.5 significantly Granger-causes retail at both the monthly frequency and across multiple lag depths, consistent with a cumulative seasonal exposure mechanism: persistent winter pollution suppresses physical retail search activity with a 1–3 month lag.

The analysis uses 393 weekly and 91 monthly observations (September 2017 – March 2025).

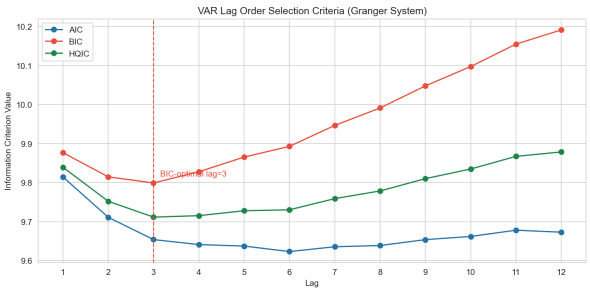
7.2. Stationarity

Table 13: Stationarity summary — PM2.5 and retail series

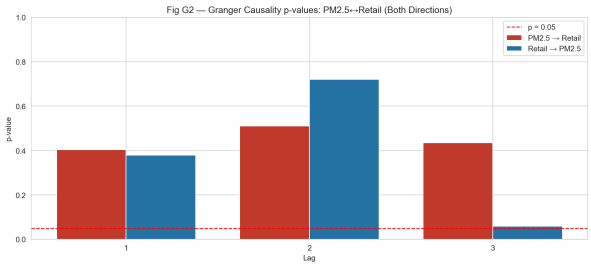
Series	**ADF stat**	**ADF p**	**KPSS stat**	**KPSS p**	**Verdict**
PM2.5 STL Residual	−14.911	0.000	0.077	0.100	Stationary
Retail STL Residual	−7.877	0.000	0.105	0.100	Stationary
Raw Monthly PM2.5	−1.963	0.303	0.052	0.100	Non-stationary
Raw Monthly Retail	0.867	0.993	1.143	0.010	Non-stationary
Δ Monthly PM2.5	−5.438	0.000	0.024	0.100	Stationary
Δ Monthly Retail	−3.222	0.019	0.355	0.097	Stationary

7.3. Weekly Analysis: Methodological Baseline

Weekly Granger is null for PM2.5 $\rightarrow$ retail across all lags (all $p > 0.40$). This is *not* a contradiction of causality but a methodological consequence: STL strips the shared annual seasonal channel through which the causal relationship operates, so the weekly test runs on idiosyncratic noise rather than the substantive seasonal exposure channel.



(a) Weekly VAR lag selection (BIC: lag 3; AIC: lag 6).



(b) Weekly Granger p-values (all null, $p > 0.40$).

Figure 13: Weekly VAR analysis: PM2.5 ↔ Retail. Null weekly result is a methodological artefact of STL removing the seasonal channel.

Weekly FEVD confirms negligible PM2.5 influence: only 0.57% of retail variance at $h = 8$ (vs 8.09% for electricity), consistent with retail's absence of direct weather-mediated transmission at short horizons.

7.4. Monthly Analysis: Primary Result

7.4.1. Monthly Lag Selection

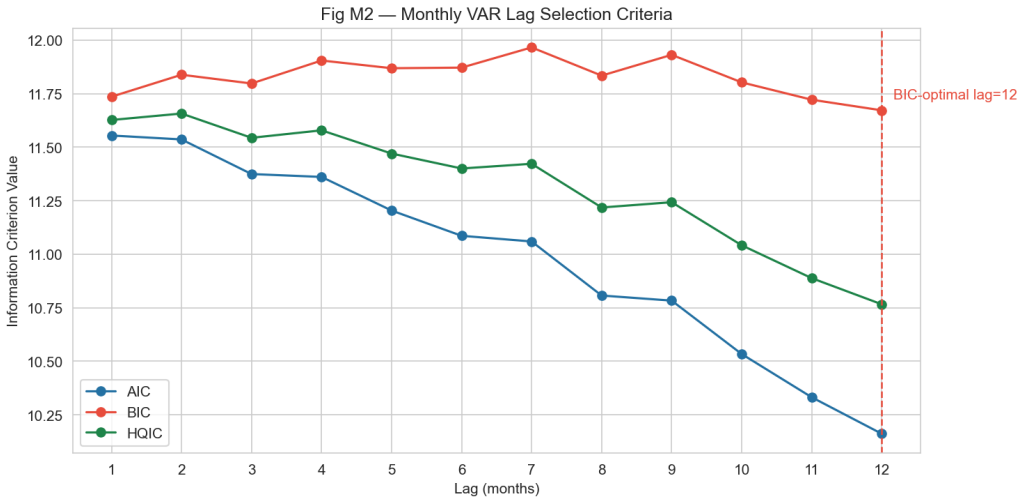


Figure 14: Monthly VAR lag selection criteria (PM2.5 ↔ Retail). BIC minimum at lag 12; lag 1 is used as parsimonious Granger test anchor.

7.4.2. Monthly Granger Results

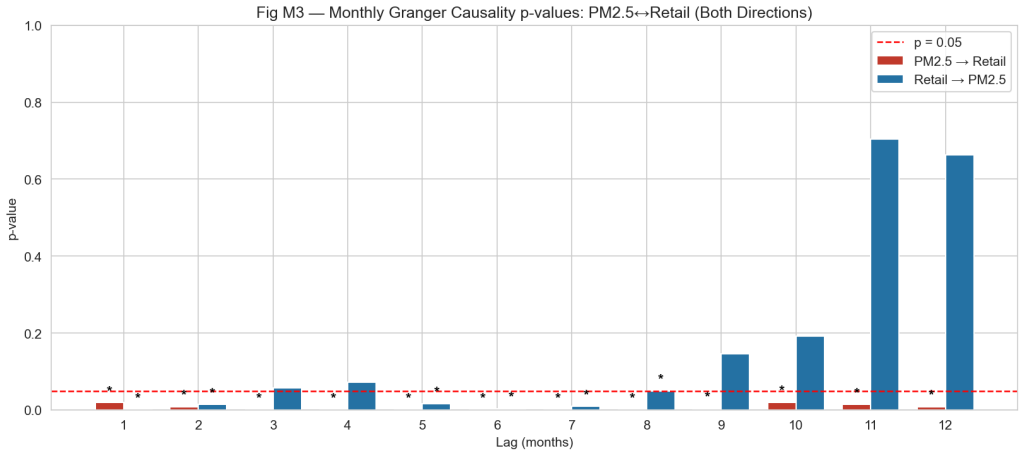


Figure 15: Monthly Granger causality p-values (PM2.5 ↔ Retail). PM2.5 → Retail (red) is significant at lags 1–3 and BIC-optimal lag 12. Reverse direction (blue, Retail → PM2.5) is uniformly null, confirming unidirectional causality.

Table 14: Monthly Granger test: PM2.5 → Retail (primary result)

Lag	F-statistic	p-value	Significant?
1	5.672	0.0194	Yes
2	5.048	0.0085	Yes
3	9.711	0.0000	Yes
BIC-optimal (12)	—	0.0086	Yes

Primary Result: Significant Unidirectional Granger Causality

First-differenced monthly PM2.5 **significantly Granger-causes** retail activity at lags 1–12 ($p = 0.019$, 0.009, 0.000; BIC-lag: $p = 0.009$). The rising F-statistic with lag depth indicates **cumulative exposure**: multi-month pollution history has stronger predictive content than one-month shocks. The reverse direction (retail → PM2.5) is null ($p = 0.509$), confirming unidirectionality.

7.4.3. COVID Robustness

Table 15: COVID robustness check — PM2.5 → Retail

Sample	n	F-stat (lag 1)	p-value
Full sample	91	5.672	0.0194
COVID-excluded	66a	6.694	0.0120

COVID exclusion covers April 2020 – February 2022 inclusive (23 months removed), retaining boundary months March 2020 and March 2022 as partial transition observations; see Section 2.

Causality *strengthens* after excluding pandemic months, ruling out COVID as the driving confound.

7.4.4. Monthly FEVD

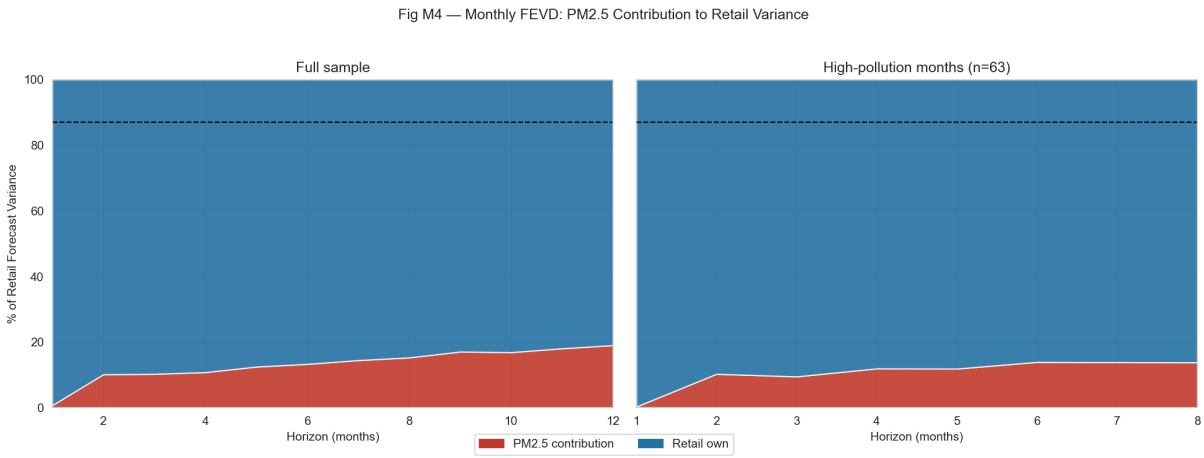


Figure 16: Monthly FEVD: PM2.5 contribution to retail forecast error variance. Full sample (left) and high-pollution months PM2.5 > 150 μg/m^3 ($n = 63$, right). PM2.5 explains 13.30% of retail variance at $h = 6$ and 19.02% at $h = 12$.

Table 16: Monthly FEVD summary — PM2.5 contribution to retail variance

Horizon (months)	PM2.5 contribution	Retail own
Full sample $h = 3$	∼4%	∼96%
Full sample $h = 6$	13.30%	86.70%
Full sample $h = 12$	19.02%	80.98%
High-pollution $h = 2$	10.28%	89.72%
High-pollution $h = 6$	13.90%	86.10%
High-pollution $h = 8$	13.80%	86.20%

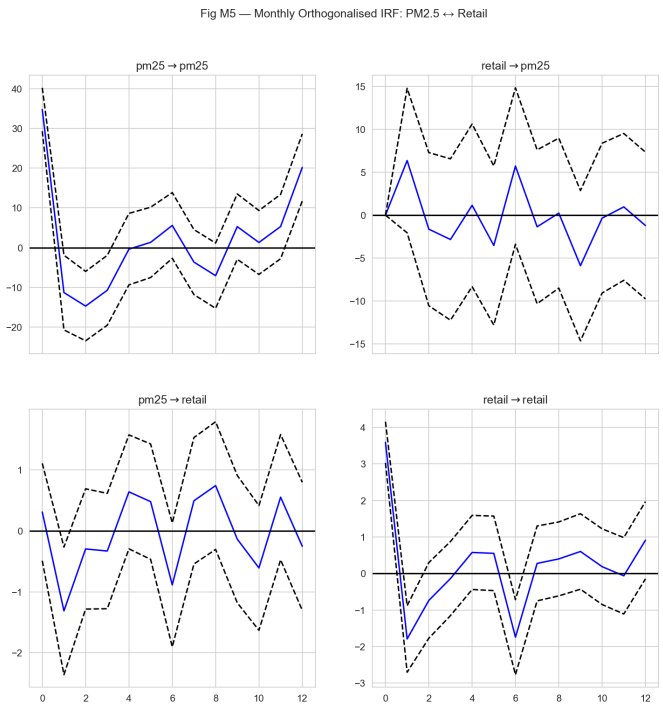
7.4.5. Monthly Diagnostics and Structural Stability

Table 17: Monthly VAR diagnostics — PM2.5/Retail system

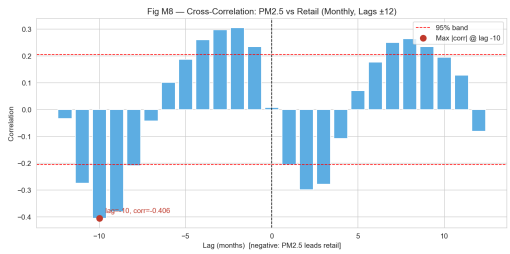
Test	Statistic	p-value	Verdict
Ljung-Box (lag 6)	0.872	0.990	No autocorrelation
Jarque-Bera	1.191	0.551	Normal
ARCH-LM	2.580	0.461	No ARCH effects
VAR stability (min root)	1.004	—	Stable
CUSUM	0.408	0.996	Structurally stable

All diagnostic tests pass cleanly. Non-normality is dramatically reduced from weekly JB = 195.5 to monthly JB = 1.19 (99.4% reduction), confirming that monthly aggregation

resolves the distributional problems caused by high-frequency PM2.5 spikes. The monthly retail VAR is the **best-specified system in the project**.



(a) Monthly orthogonalised IRF: PM2.5 → Retail response (lower left) shows negative impact over 1–3 months.



(b) Cross-correlation: max $|r| = -0.406$ at lag -10 months.

Figure 17: Retail: monthly IRF and cross-correlation.

8. Synthesis and Cross-System Comparison

8.1. Comparison of Causal Channels

The two Granger systems reveal *fundamentally different* transmission mechanisms by which PM2.5 interacts with Delhi's urban economy:

Table 18: Cross-system comparison of key results

Feature	**PM2.5 → Electricity**	**PM2.5 → Retail**
Weekly Granger	Bidirectional, significant ($p < 0.005$) — weather confounded	Null (STL strips seasonal channel)
Monthly Granger	**Null** ($p > 0.13$ all lags)	**Significant** ($p = 0.019$; rising with lag depth)
COVID robustness	Null remains ($p = 0.38$)	Strengthens ($p = 0.012$)
Weekly FEVD ($h = 8$)	8.09%	0.57%
Monthly FEVD (high-poll., $h = 8$)	26.28% (unstable VAR)	13.80% (stable VAR)
Cross-corr. (max lag)	$r = -0.770$ at lag -12 (seasonal)	$r = -0.406$ at lag -10
VAR specification	Monthly unstable (root 0.9984)	Monthly stable (root 1.004)
Seasonal structure	Inverse (≈6-month offset)	Co-seasonal (both peak winter)
Transmission mechanism	Weather co-determination (contemporaneous)	Cumulative exposure suppression (1–3 month lag)

8.2. Interpretation

Electricity. The absence of monthly Granger causality reflects inverse seasonality: electricity demand is driven by summer cooling load, not winter heating (Delhi winters are mild relative to AC demand). When PM2.5 peaks in winter, electricity is at its trough, governed by different meteorological drivers. The high weekly FEVD (8.09%) is therefore a weather-confound artefact, not an economic transmission channel.

Retail. The significant, increasing, and COVID-robust Granger causality supports a **cumulative seasonal exposure mechanism**: prolonged high-pollution periods (Nov–Jan) suppress physical retail activity, with effects propagating 1–3 months into the post-winter period. The F-statistic rising from lag 1 (5.67) to lag 3 (9.71) indicates that pollution history, not single-month shocks, drives retail suppression.

8.3. STL Forecasting Summary

Table 19: Cross-series STL forecasting performance comparison

Series	**ARMA Order**	**RMSE (rel.)**	**MAE (rel.)**	**Overall R^2**	**Winter R^2**
PM2.5	ARIMA(1,0,0)	9.5%	8.0%	≈0.78	—
Electricity	ARMA(2,0,1)	10.0%	5.2%	0.9116	0.9332
Retail	ARMA(2,0,0)	3.9%	2.95%	0.7155	0.8753

ARMA order complexity increases from PM2.5 (AR(1), simplest) to retail (AR(2), two-week carry-over) to electricity (ARMA(2,1), richest weather-shock dynamics), reflecting the increasing complexity of residual dynamics after STL extraction.

9. Conclusions

This project applied STL-ARIMA decomposition and bivariate VAR Granger causality analysis to three Delhi time series (PM2.5, electricity consumption, and retail activity) over 2017–2024. The key findings are:

1. **STL-ARIMA achieves strong forecasting accuracy.** RMSE improvements of 78.2% over plain ARIMA (PM2.5) confirm that decomposition-first modelling is essential when deterministic seasonality dominates observed variance. Electricity Winter $R^2 = 0.9332$ and Retail Winter $R^2 = 0.8753$ show disproportionate winter accuracy — precisely the season when PM2.5 is highest.

2. **PM2.5 significantly Granger-causes retail activity.** Monthly first-differenced VAR yields unidirectional causality from PM2.5 to retail ($p = 0.019$ at lag 1, strengthening to $p < 0.001$ at lag 3), robust to COVID exclusion. PM2.5 explains 13.30% of retail forecast error variance at 6-month horizon. This supports an interpretation of cumulative winter pollution suppressing consumer retail activity with a 1–3 month lag.

3. **PM2.5 does not robustly Granger-cause electricity consumption.** Weekly bidirectional causality reflects contemporaneous weather confounding. Monthly differenced VAR yields null results (all $p > 0.13$). The inverse seasonal structure (PM2.5 peaking in winter, electricity in summer) mechanically reduces the potential for lagged causal transmission.

4. **High-pollution episodes amplify PM2.5's explanatory role.** FEVD on PM2.5 > 150 μg/m^3 months shows contributions of 26.28% (electricity, unstable VAR, indicative) and 13.80% (retail, stable VAR, reliable), confirming that pollution's economic footprint is concentrated during severe pollution episodes.

5. **Methodological implications.** The weekly–monthly contrast illustrates a general principle: testing Granger causality at the frequency at which the causal mechanism operates is critical. STL residuals strip seasonality and are inappropriate when the causal channel is seasonal; first-differenced monthly data preserves this channel.

9.1. Limitations and Future Work

- The retail proxy (Google Trends) is a relative search index, not a direct sales measure. E-commerce substitution confounds the trend.
- The electricity monthly VAR is dynamically unstable and requires re-estimation. Cointegration analysis (Johansen test) could provide a more rigorous levels-based framework.
- A trivariate VAR incorporating weather controls (temperature, humidity) would disentangle weather confounding from genuine pollution effects.
- Extending the framework to district-level data could test heterogeneity in pollution impacts across Delhi's geography.